# Reuben Seyram Komla Agbozo

*Human Factors Researcher · User Behavior Architect*

Problem solving · Human–machine interaction · Research and design

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## Profile

I am a problem solver and user behavior architect focused on how people think, act and adapt within systems. I notice what isn't working for people, define the real problem, and build practical solutions: products, prototypes, learning programs and the way an organisation runs. I have founded and led a team from scratch, designed AI curricula and trained teachers to use AI, and developed a company's organisational structure and standardised ways of measuring its goals. I work across engineering, design and education, and across cultures in English and Mandarin, turning complex ideas into things people can use.

## Core strengths

| Strength | What I bring |
|---|---|
| **Problem framing** | Turns vague, messy problems into clear questions a team can answer and act on. |
| **Human behavior and systems** | Designs systems that adapt to how people actually behave, instead of asking people to adapt to the system. |
| **Building and testing** | Builds working prototypes quickly, so ideas are tested before time and money are committed. |
| **Analytical judgment** | Combines hard data with what people report to reach conclusions that hold up under scrutiny. |
| **Learning new fields fast** | Becomes productive quickly in unfamiliar fields and carries ideas from one domain into another. |
| **Critical review** | Gives clear, fair assessments that help others improve their work. |
| **Auditing and quality systems** | Checks whether processes deliver what they promise and finds where they fall short. |
| **AI learning and communication** | Helps people and teams adopt AI, and explains technical ideas in terms anyone can use. |
| **Leadership and organisation** | Builds teams and structures that make work clear, measurable and efficient. |

## Methods and tools

| Area | Experience |
|---|---|
| **Study design and analysis** | Within-subject experiments; counterbalancing; repeated-measures and mixed-effects analysis; correlation, multiple-testing correction and sensitivity analysis. |
| **Human-factors measurement** | Tobii eye tracking; Xsens motion capture; NASA-TLX; synchronized multimodal data processing and interpretation. |
| **Simulation and computation** | CARLA; CarSim; model predictive control (MPC); Logitech steering and pedal interfaces; Python analysis pipelines; vehicle telemetry; machine vision with Mask R-CNN; Neo4j knowledge graphs. |
| **AI and educational design** | AI-assisted prototyping; web-app and agent workflows; curriculum development; assessment design; teacher training. |

## Research and professional experience

| Period | Role | Organisation | Contribution | Outcomes |
|---|---|---|---|---|
| Sep 2021 to present | **Doctoral Researcher, Human–Intelligent Vehicle Interaction** *Jan 2024 to present* | Zhejiang University, Hangzhou | • Built the group's first driving-simulation lab, including a driver assistance system with steering, pedal and combined support.<br>• Developed the Driver Workload Assessment Methodology (DWAM), which combines visual, physical and reported workload to show whether, where and in which direction workload changes when driving tasks shift to the system.<br>• Guided two master's students in designing and building their experiments and analysing their data to answer their research questions. | • The lab now supports the group's ongoing and future studies.<br>• Introduced driver workload into the SAE levels of driving automation, which define only what the driver must do, together with a methodology to quantify it. In 51 drivers, every support mode reduced workload, each in a different part of the body.<br>• Both master's students graduated successfully and are now employed. |
| | **Doctoral Researcher, Smart Sustainable Manufacturing** *Sep 2021 to Dec 2023 Including Research Assistant with Siemens, Sep 2022 to Aug 2023* | Zhejiang University, Hangzhou Siemens, Shanghai | • Proposed the concept of cognitive intelligence-enabled manufacturing.<br>• Investigated causality-based quality problem solving for discrete manufacturing systems.<br>• With Siemens: built a data-driven Neo4j knowledge graph of low-carbon manufacturing processes to support automated life cycle assessment.<br>• With Siemens: studied how big data can support greener, smarter production across the production value chain. | • First-author paper, "Towards Cognitive Intelligence-Enabled Manufacturing," APMS 2022.<br>• Co-authored "Two-stage approach to causality analysis-based quality problem solving for discrete manufacturing systems": Best Student Conference Paper, NSFC-RGC 2023 Conference, Hong Kong. Extended version published in the Journal of Engineering Design (2025).<br>• Co-authored "Knowledge graph-based mapping and recommendation to automate life cycle assessment," Advanced Engineering Informatics (2024).<br>• First-author perspective, "Enhancing big data for greentelligence across the production value chain," Green Manufacturing Open (2023). |
| Apr 2024 to present *Freelance* | **Program Research Lead and Advisor** *Apr 2024 to present* | Emerald Hill Design Education Institute, Hangzhou | • Designed the institute's curriculum on AI, web apps and AI agents.<br>• Helped teachers bring AI into their teaching.<br>• Advised leadership on research and management. | • Institute revenue increased by 18%.<br>• About 30 students and working professionals trained. |
| | **Management Consultant** *Nov 2024* | Komodo Mining Services, Ghana | • Restructured the company's service offerings into a clear, marketable set of services.<br>• Designed its people management system, including how work is reported and quantified. | • The new service structure helped the company secure overseas contracts.<br>• Consistent reporting and measurable work output across the team. |
| | **AI and Robotics Curriculum Consultant** *Jun to Aug 2025* | Belio Technology (a subsidiary of WaShalimba Group), Namibia | Designed progressive AI and robotics curricula for four learner groups: ages 9 to 11, 12 to 15 and 16 to 18, and adult and professional learners. | |
| Aug 2020 to Aug 2021 | **Research Assistant** | Lanzhou University of Technology, Lanzhou | Developed a Mask R-CNN method to locate, segment and measure carbide particles in steel microscopy images, replacing manual image-by-image analysis. | • Over 90% of target particles located and segmented.<br>• First-author paper, Journal of the Korean Society for Precision Engineering (2020). |

## Leadership and teaching experience

| Period | Role | Organisation | Contribution | Outcomes |
|---|---|---|---|---|
| Mar 2018 to Aug 2021 | **Founding Editor in Chief** | School of International Education, Lanzhou University of Technology | • Founded and led the school's first editorial and media team.<br>• Created a website, YouTube channel, Facebook and Instagram pages, promotional materials and an overseas partnership program. | • Team of about 10 built from scratch.<br>• Four online channels launched to support international student admissions. |
| Aug 2015 to Aug 2016 | **Teaching Assistant** | University of Ghana, Accra | Taught engineering students as a lecturer's assistant in algebra, calculus 1 and 2, environmental engineering, and thermodynamics 1 and 2. | • Six courses.<br>• Classes of up to about 200 students. |
| 2009 to 2011 | **Science and Mathematics Teacher** | Stage International School, Ashaiman, Ghana | Taught science and mathematics to primary and middle school students. | Every student passed the Basic Education Certificate Examination (BECE) and gained admission to senior high school. |

## Education

| Period | Qualification | Institution | Focus and distinction |
|---|---|---|---|
| Sep 2021 to Mar 2027 (expected) | PhD, Industrial and Systems Engineering (Mechanical Engineering) | Zhejiang University | Driver workload assessment and human well-being in smart vehicles. |
| Sep 2017 to Jun 2020 | MEng, Mechanical Manufacture and Automation | Lanzhou University of Technology | Digital image processing for quantitative metallography. Best Graduating Student and Valedictorian. |
| Sep 2016 to Jun 2017 | Chinese Language and Culture | Lanzhou University of Technology | HSK levels 4 and 5. |
| Sep 2011 to Jun 2015 | BSc Engineering, Process Engineering | University of Ghana | Process plant design. |

## Publications and writing

Publication record: ORCID 0000-0001-8147-7609. Selected publications are noted alongside the relevant roles.

Author of "The Tyranny of the Default: Toward a User Behavior Architecture," exploring how systems could adapt to sensed human behavior rather than impose a fixed default. Publishes problem-solving commentary on LinkedIn as "Reuben ASK: Insights & Opinions."

## Certifications

| Year | Certification | Issuer |
|---|---|---|
| 2020 | Internal Auditor: ISO 9001:2015, ISO 14001:2015 and ISO 45001:2018, covering quality, environmental, and occupational health and safety management. | BSI Training Academy, Singapore |
| 2018 | Advanced TESOL Certification and Teaching English to Children. | American TESOL Institute, Shanghai |

## Awards and recognition

| Year | Award |
|---|---|
| 2023 | Best Student Conference Paper, 2023 NSFC-RGC Conference on Frontiers of Industrial Big Data and Intelligent Systems, Hong Kong. |
| 2020 | Best Graduating Student and Valedictorian, Lanzhou University of Technology. |
| 2019 | Chinese Government Outstanding International Student Award, Ministry of Education of China: one of 582 international students selected from 127 universities nationwide. |

## International memberships and organisations

| Organisation | Membership |
|---|---|
| Human Factors and Ergonomics Society (HFES) | Member (Emerging Country Membership) |

## Languages

English (native) · Ewe · Twi · Ghanaian Pidgin English · Mandarin Chinese (professional, HSK 5) · French (elementary)

References available on request.